

AI-POWERED SMART TRAFFIC CONGESTION PREDICTOR FOR SMART CITIES

PREDICT TRAFFIC BEFORE IT HAPPENS

An intelligent system that predicts traffic congestion **30–60 minutes in advance** and helps authorities and commuters take preventive action.



Save Time



Reduce Emissions



Faster
Emergency
Response



Smarter
Mobility



PROBLEM

Traffic congestion causes huge delays, fuel wastage, pollution and affects productivity. Current systems react after congestion occurs.

We need to predict and prevent it!



OBJECTIVES

- ✓ Predict congestion 30–60 minutes in advance
- ✓ Identify high-risk congestion zones
- ✓ Suggest alternative routes
- ✓ Optimize traffic signal timing (Advanced)
- ✓ Provide real-time alerts to commuters



INPUT DATA SOURCES

- Historical Traffic Data
- Live GPS / Location Data
- Weather Conditions
- Festival / Event Schedules
- Accident Reports
- Public Transport Density
- CCTV / Video Feed (Bonus)



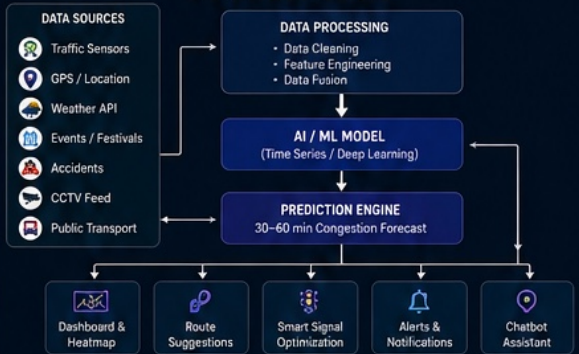
SOLUTION FEATURES

- ✓ Traffic Heatmap of Congested Roads
- ✓ AI Congestion Prediction Dashboard
- ✓ Smart Rerouting Suggestions
- ✓ Emergency Vehicle Priority Route
- ✓ Real-time Traffic Alerts
- ✓ AI Chatbot for Traffic Updates
- ✓ Mobile / Web Dashboard

HOW IT WORKS



SYSTEM ARCHITECTURE



DASHBOARD PREVIEW



Heatmap



Alternative
Routes



Signal
Timing



Emergency
Route



Live
Alerts



INNOVATION & PREVENTIVE ACTIONS

- Dynamic Signal Timing based on real-time prediction
- Carpool Suggestions to reduce vehicle count
- Smart Parking Diversion to avoid parking congestion
- Event-based Traffic Prediction (festivals, concerts, matches)
- Green Route Suggestions to reduce pollution



TECH STACK

- AI / ML
Python, TensorFlow, Scikit-learn, LSTM, XGBoost
- FRONTEND
React.js / Flutter
- BACKEND
Python Flask / Node.js
- MAPS & VISUALIZATION
Google Maps API / OpenStreetMap
- DATABASE
Firebase / MongoDB / PostgreSQL



DELIVERABLES (22 HOURS)

- Working Prototype
- AI Prediction Model
- Dashboard / UI
- Real-time Alerts & Notifications
- Demo Presentation with Real Scenario Testing



EVALUATION CRITERIA

Innovation & Creativity	25%
AI Model Accuracy	20%
Technical Implementation	20%
UI / UX & Dashboard	15%
Real-world Impact	10%
Final Presentation	10%
TOTAL	100%

SMARTER ROADS. SMARTER CITIES. BETTER TOMORROW.

#PredictPreventProceed

Let's build the future of
intelligent urban mobility!

Innovation Feature: Smart Route with Toll Fee Estimation

New Feature (NHAI Toll Intelligence):

- Show estimated toll fee on each route as per NHAI/FASTag data
- Compare Fastest Route vs Lowest Toll Route vs AI Recommended Route
- Display ETA, Toll Cost, Traffic Level and Fuel Efficiency
- Suggest Toll-Free Alternatives for commuters

Example:

Route A – Fastest Route

Time: 48 mins | Toll Fee: ₹135 | Traffic: Medium

Route B – Economical Route

Time: 62 mins | Toll Fee: ₹0 | Traffic: Low

Route C – AI Recommended

Time: 52 mins | Toll Fee: ₹60 | Lowest Congestion

Benefits:

- Better commuter decision making
- Cost optimization
- Smarter route planning
- Real-world smart city impact